In-Class Assignment L7-2 (Group of 3 or less): Survey Design

(25 minutes) Submit to LEB2

The graph below represents the purchase histories of 4 consumers buying 5 products. An edge between consumer a_i and product b_j indicates that consumer a_i has bought product b_j. The following constraints are assumed:

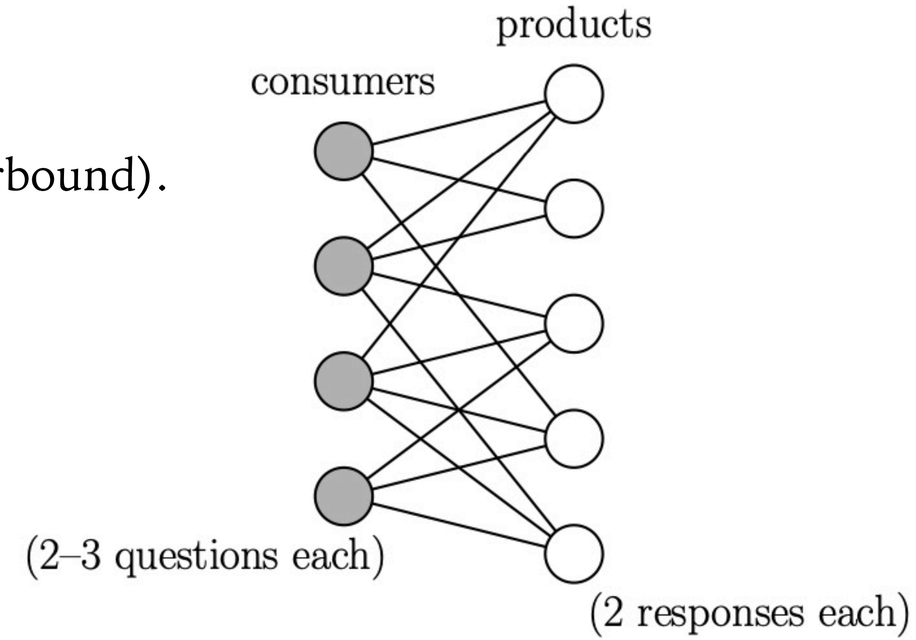
- Every consumer should be asked at least two questions and no more than three questions.
- For each product, we want exactly two replies from consumers.

(a) Draw the circulation network with lowerbounds.

(b) Draw the circulation network with lowerbounds eliminated (zero lowerbound).

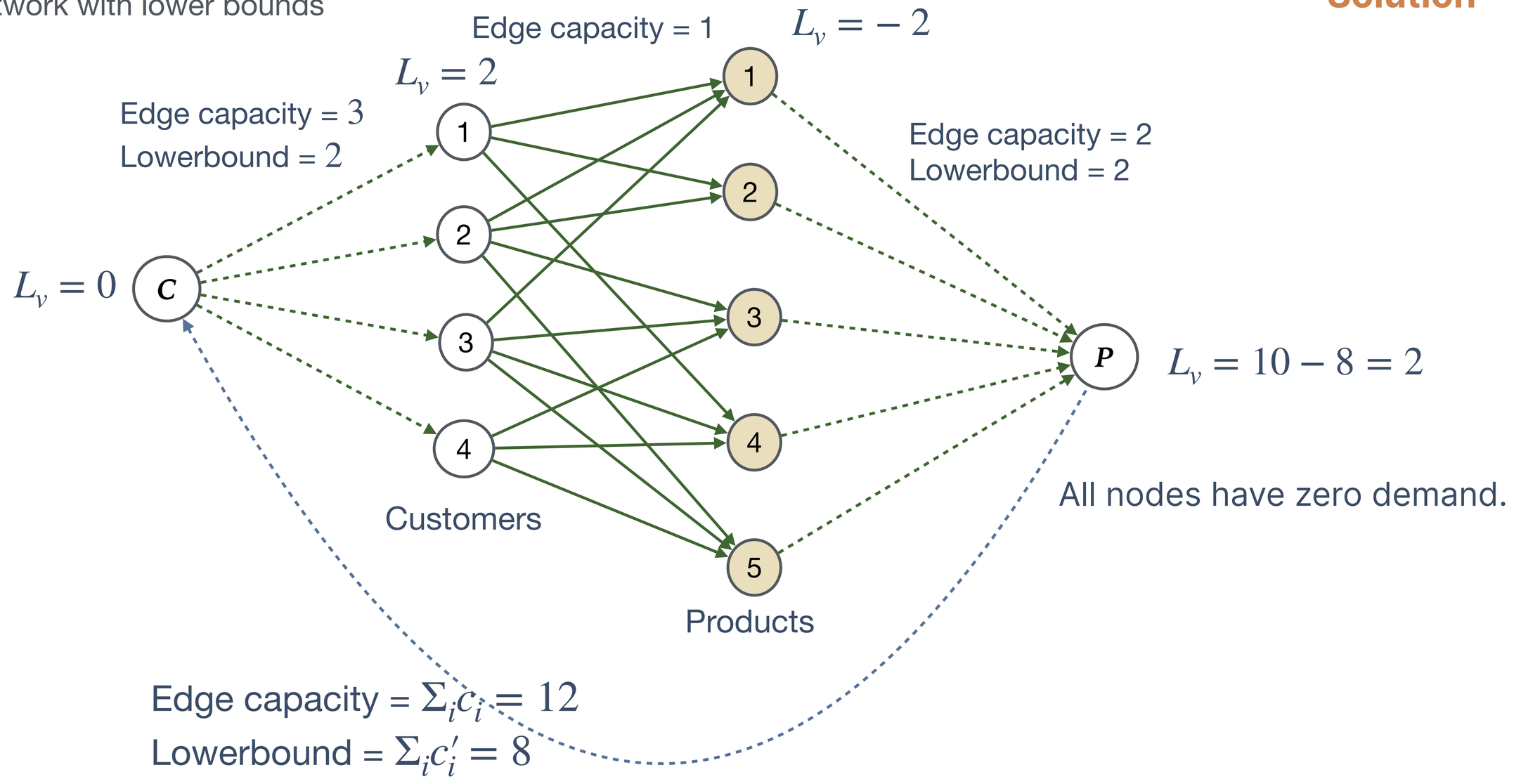
(c) Draw the flow network that is used to find a solution to this survey design problem.

(d) [Optional] Solve for the solution from the flow network and show the final survey design.



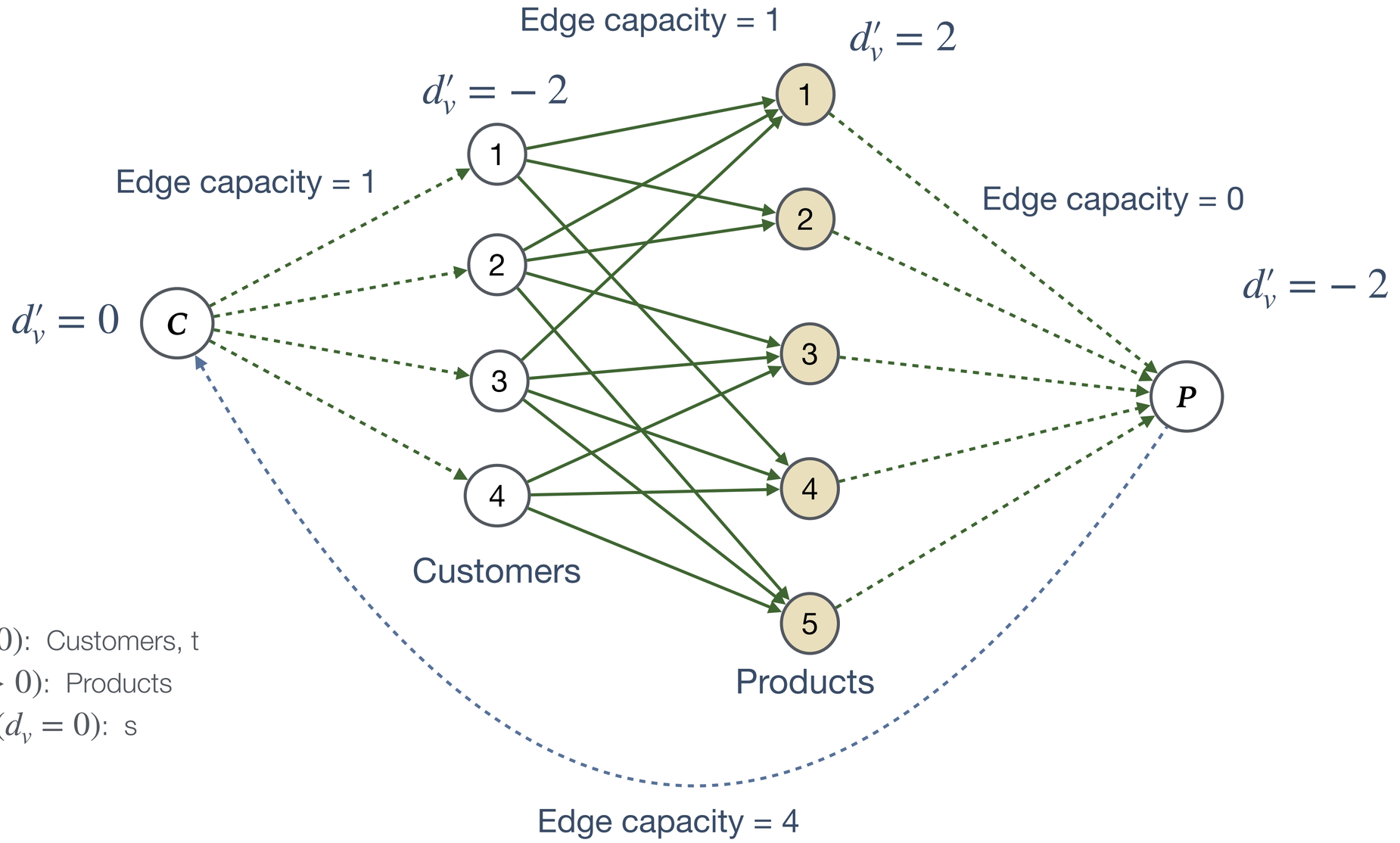
Circulation network with lower bounds

Solution



Circulation network with no lower bounds

Solution



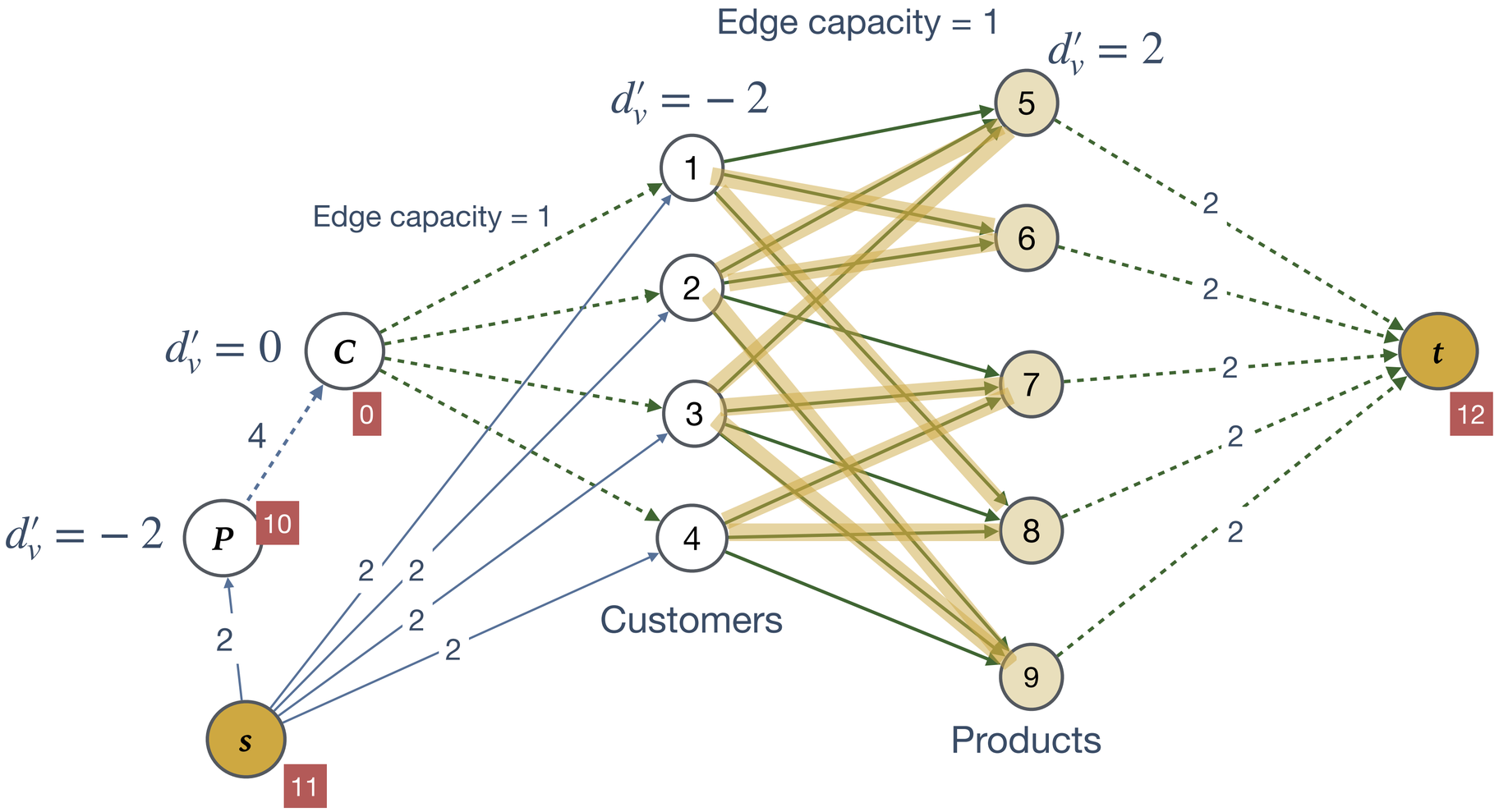
Supply nodes ($d_v < 0$): Customers, t

Demand nodes ($d_v > 0$): Products

Transhipment nodes ($d_v = 0$): s

Flow network obtained from the circulation network with no lower bounds.

Solution

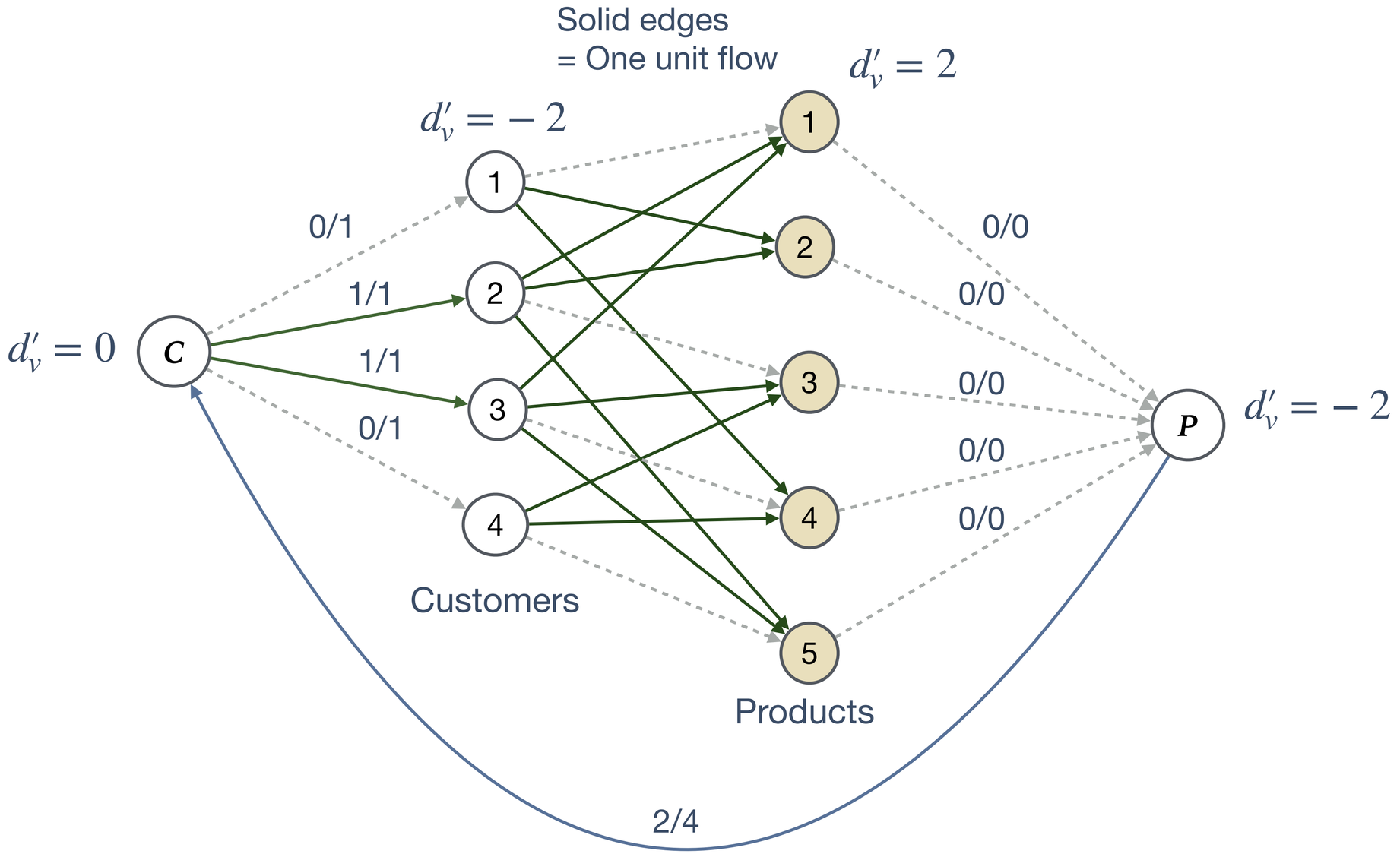



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Max flow: 10

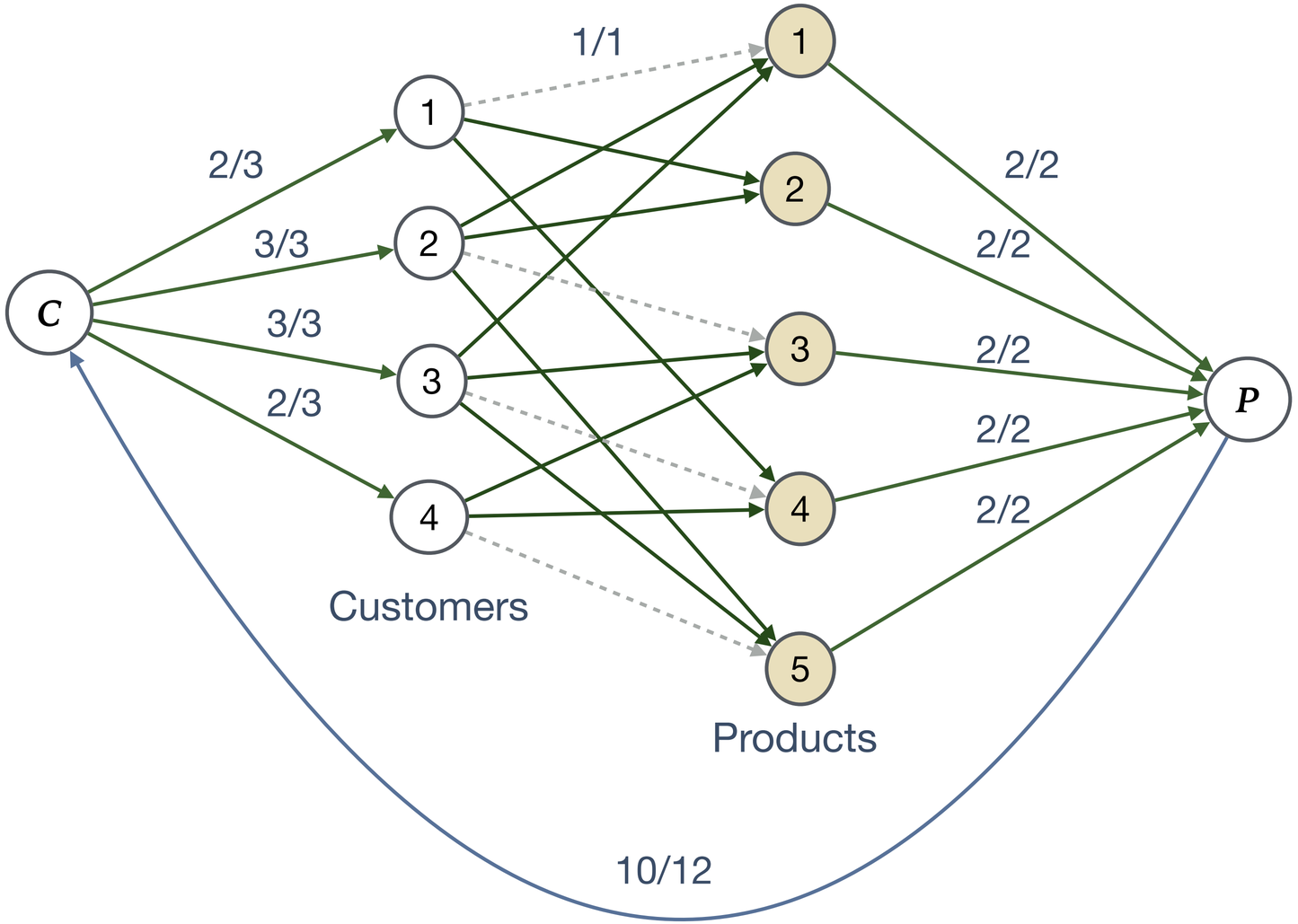
  Arc    Flow / Capacity
11 / 1     2  /   2
11 / 2     2  /   2
11 / 3     2  /   2
11 / 4     2  /   2
11 / 10    2  /   2
1 / 5      0  /   1
1 / 6      1  /   1
1 / 8      1  /   1
2 / 5      1  /   1
2 / 6      1  /   1
2 / 7      0  /   1
2 / 9      1  /   1
3 / 5      1  /   1
3 / 7      1  /   1
3 / 8      0  /   1
3 / 9      1  /   1
4 / 7      1  /   1
4 / 8      1  /   1
4 / 9      0  /   1
10 / 0     2  /   4
0 / 1      0  /   1
0 / 2      1  /   1
0 / 3      1  /   1
0 / 4      0  /   1
5 / 12     2  /   2
6 / 12     2  /   2
7 / 12     2  /   2
8 / 12     2  /   2
9 / 12     2  /   2
Source side min-cut: [11]
Sink side min-cut: [12]
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Solution in circulation network with no lower bounds

Solution

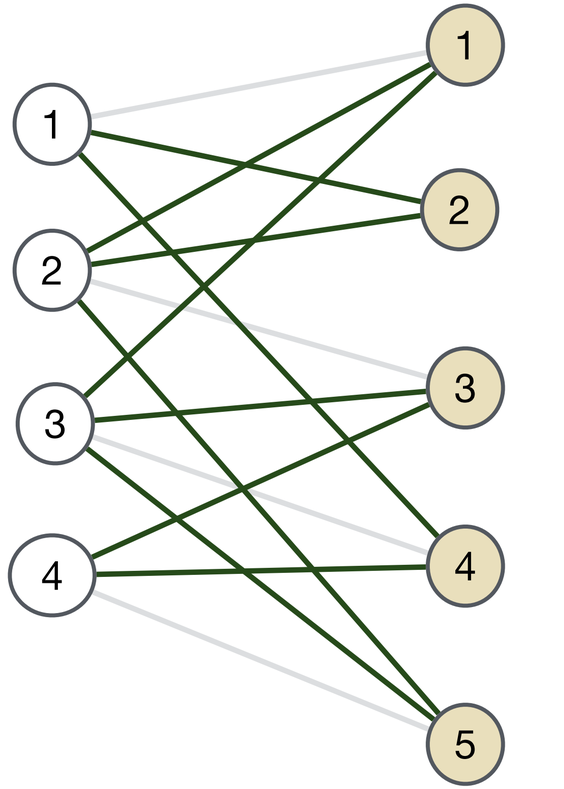


Solution in circulation network with lower bounds



Solution

Final survey



All nodes have zero demand.
Feasible circulation with lowerbound